METRIC

FF-B-171/12

30 NOVEMBER 1993

FEDERAL SPECIFICATION SHEET

# BEARING, BALL, ANNULAR, DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING, ANSI/AFBMA DIMENSION SERIES 03.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

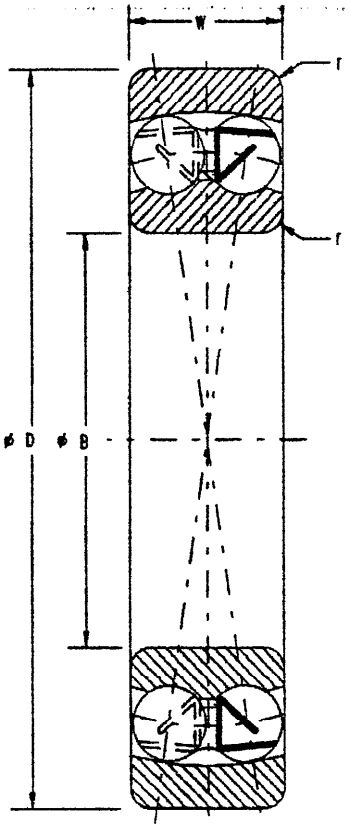



FIGURE 1.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH CYLINDRICAL BORE (CODE C)

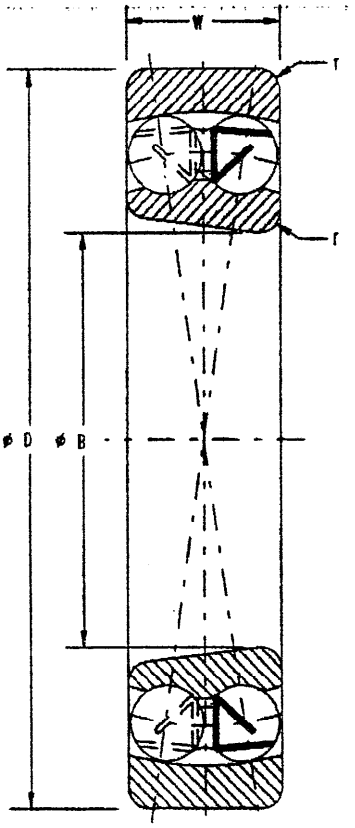



FIGURE 2.
DOUBLE ROW, RADIAL, INTERNAL SELF-ALIGNING WITH TAPERED BORE (CODE T)

## BEARING CONFIGURATIONS

AMSC/NA FSC 3110

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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Design. All self-aligning bearings shall be open. Bores shall be cylindrical or tapered (taper 1:12) and shall be specified in the part number by the codes in table III.

4. Misalignment. Maximum permissible misalignment shall be 4.0 degrees.

5. Radial Internal Clearante. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

6. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

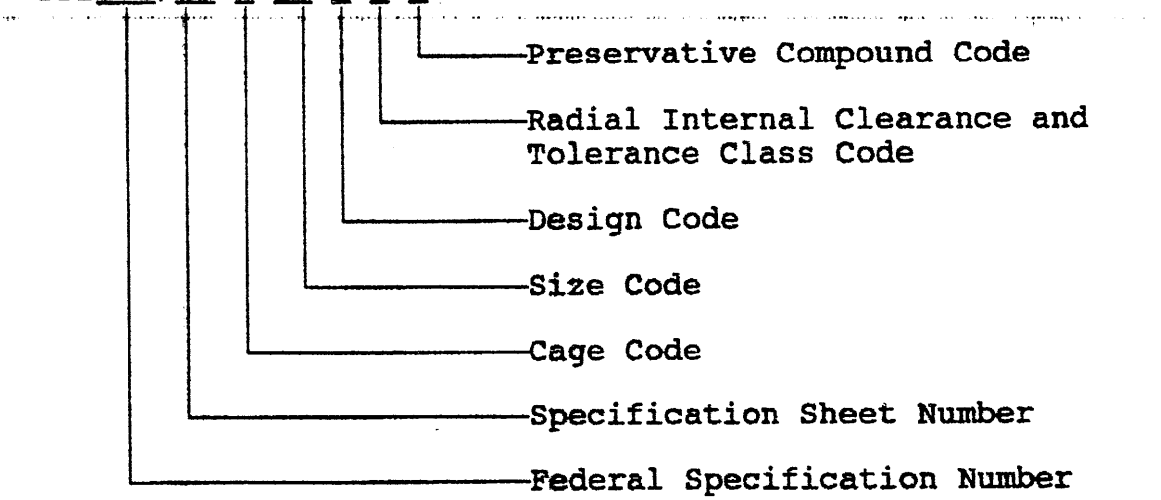
7. Preservative Compound. All bearings shall be supplied with preservative compound in accordance with MIL-P-197 and designated by the code “C” in the part number.



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8. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, bore design, radial internal clearance and tolerance class, and preservative compound.

EXAMPLE: FFB171/12 M 20 T 3 C




FFB171/12M20T3C indicates - Bearing, cage-machined bronze or brass; bore 100mm; outside diameter 215mm; width 47mm; open (no shields or seals), tapered bore; radial internal clearance symbol 3 (greater than normal); tolerance class ABEC-1; preservative compound in accordance with MIL-P-197.



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TABLE I. Boundary dimensions.

| Size Code 1/ | OB Bore | OD Outside Diameter | W Width | Chamfer $r_s$min 2/ | Shaft Shoulder Diameter Min 3/ | C Dynamic load rating (pounds) |
|---|---|---|---|---|---|---|
| 00 | 10 | 35 | 11 | 0.6 | 15 | 1250 |
| 01 | 12 | 37 | 12 | 1.0 | 18 | 1605 |
| 02 | 15 | 42 | 13 | 1.0 | 21 | 1675 |
| 03 | 17 | 47 | 14 | 1.0 | 23 | 2163 |
| 04 | 20 | 52 | 15 | 1.1 | 27 | 2145 |
| 05 | 25 | 62 | 17 | 1.1 | 32 | 3100 |
| 06 | 30 | 72 | 19 | 1.1 | 37 | 3690 |
| 07 | 35 | 80 | 21 | 1.5 | 43.5 | 4230 |
| 08 | 40 | 90 | 23 | 1.5 | 48.5 | 5120 |
| 09 | 45 | 100 | 25 | 1.5 | 53.5 | 6590 |
| 10 | 50 | 110 | 27 | 2.0 | 60 | 7420 |
| 11 | 55 | 120 | 29 | 2.0 | 65 | 8885 |
| 12 | 60 | 130 | 31 | 2.1 | 72 | 9920 |
| 13 | 65 | 140 | 33 | 2.1 | 77 | 10850 |
| 14 | 70 | 150 | 35 | 2.1 | 82 | 12795 |
| 15 | 75 | 160 | 37 | 2.1 | 87 | 13680 |
| 16 | 80 | 170 | 39 | 2.1 | 92 | 15300 |
| 17 | 85 | 180 | 41 | 3.0 | 99 | 16975 |
| 18 | 90 | 190 | 43 | 3.0 | 104 | 20525 |
| 19 | 95 | 200 | 45 | 3.0 | 109 | 23535 |
| 20 | 100 | 215 | 47 | 3.0 | 114 | 24680 |
| 21 | 105 | 225 | 49 | 3.0 | 119 | 26450 |
| 22 | 110 | 240 | 50 | 3.0 | 124 | 27910 |
| 24 | 120 | 260 | 55 | 3.0 | 134 | 33265 |
| 26 | 130 | 280 | 58 | 4.0 | 148 | 37015 |
| 28 | 140 | 300 | 62 | 4.0 | 158 | 42080 |

1/ Size codes in Table I are industry bore codes.

2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.

3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.



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TABLE II. Codes for Cages.

| Code | Cage |
|---|---|
| J | Pressed steel or molded plastic |
| M | Machined bronze or brass |
| T | Machined non-metallic (phenolic) |

TABLE III. Design Codes.

| Code | Design |
|---|---|
| C | Open (no shields or seals), Cylindrical bore - fig.1 |
| T | Open (no shields or seals), Tapered bore - fig.2 |

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

| Code | Radial Internal Clearance | Tolerance Class |
|---|---|---|
| 1 | Symbol 2 (less than normal) | ABEC-1 |
| 2 | Symbol 0 (normal) | |
| 3 | Symbol 3 (greater than normal) | |
| 4 | Symbol 4 (greater than symbol 3) | |
| 5 | Symbol 2 (less than normal) | ABEC-5 |
| 6 | Symbol 0 (normal) | |
| 7 | Symbol 3 (greater than normal) | |
| 8 | Symbol 4 (greater than symbol 3) | |

MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army-AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - 0S

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

[Project 3110-0848-12)



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